

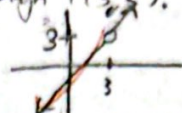
Total
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1. A function has a hole at the point $P(3,3)$, and $f(3)=4$:
- Determine $\lim_{x \rightarrow 3} f(x)$, assuming it exists.
 - Explain your reasoning. Include a rough diagram.

i) $\lim_{x \rightarrow 3} f(x) = 3$

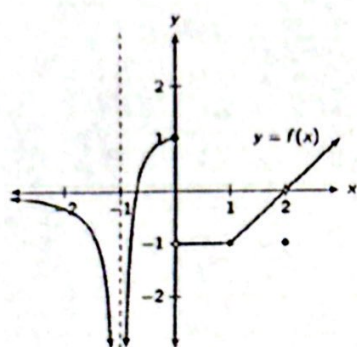
ii) Assuming that there exists a hole at $P(3,3)$ on the graph, $\lim_{x \rightarrow 3} f(x) = 3$.

This is because we can assume that without the hole, the graph would pass through $P(3,3)$. The info $f(3)=4$ does not affect the limit in this case.



Also, we know that $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$ because $\lim_{x \rightarrow 3} f(x)$ exists.

2. Use the following graph to answer the questions below:



a) $\lim_{x \rightarrow -1^-} f(x) = -\infty \therefore \text{DNE}$

b) $\lim_{x \rightarrow 0} f(x) = \text{DNE}$ since $\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$

c) $\lim_{x \rightarrow 2} f(x) = 0$

d) Is the function continuous at $x=2$? Justify.

Function has point discontinuity at $x=2$ because $\lim_{x \rightarrow 2} f(x) \neq f(2)$.

3. Evaluate the following limits. Show all work, and use proper form.

a) $\lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{6}{x^2-9} \right)$: disc 4

$= \lim_{x \rightarrow 3} \left[\frac{1}{x-3} - \frac{6}{(x-3)(x+3)} \right] = \frac{1}{(3)+3}$

$= \lim_{x \rightarrow 3} \left[\frac{(x+3)-6}{(x-3)(x+3)} \right]$

$= \lim_{x \rightarrow 3} \left[\frac{x-3}{(x-3)(x+3)} \right]$

$= \lim_{x \rightarrow 3} \left(\frac{1}{x+3} \right)$

b) $\lim_{x \rightarrow 3} \frac{|x-3|}{x-3}$

piecewise function

$f(x) = \begin{cases} \frac{x-3}{x-3} = 1, & x > 3 \\ \frac{x-3}{x-3} = -1, & x < 3 \end{cases}$

$\lim_{x \rightarrow 3^-} f(x) = -1$

$\lim_{x \rightarrow 3^+} f(x) = 1$

$\lim_{x \rightarrow 3^-} f(x) \neq \lim_{x \rightarrow 3^+} f(x)$

$\therefore \lim_{x \rightarrow 3} f(x) = \text{DNE}$

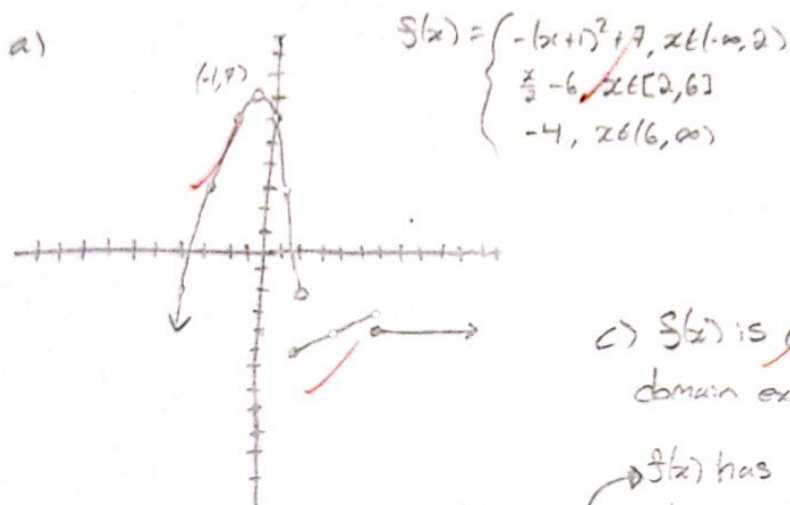
4. Given $\lim_{x \rightarrow -3} f(x) = 4$, simplify the following using properties of limits. Then, evaluate.

$$\begin{aligned} &= \frac{\lim_{x \rightarrow -3} [5(x) - 4x]}{\lim_{x \rightarrow -3} [5(x)^2]} \\ &= \frac{\lim_{x \rightarrow -3} [5(x)] - \lim_{x \rightarrow -3} [4x]}{\lim_{x \rightarrow -3} [5(x)^2]} \end{aligned} \quad \begin{aligned} &= \frac{\lim_{x \rightarrow -3} \frac{f(x) - 4x}{[f(x)]^2}}{\lim_{x \rightarrow -3} [5(x)]^2} \\ &= \frac{\lim_{x \rightarrow -3} [5(x)] - 4 \lim_{x \rightarrow -3} [x]}{[\lim_{x \rightarrow -3} 5(x)]^2} \\ &= \frac{(4) - 4(-3)}{(4)^2} = \frac{4+12}{16} = \frac{16}{16} = 1 \end{aligned}$$

5. Given the function below:

$$f(x) = \begin{cases} -(x+1)^2 + 7, & x < 2 \\ \frac{x}{2} - 6, & 2 \leq x \leq 6 \\ -4, & x > 6 \end{cases}$$

- a) Provide a sketch of the function.
b) Find $\lim_{x \rightarrow 2} f(x)$.
c) Describe the functions continuity throughout it's domain. Use an algebraic technique.



b) $\lim_{x \rightarrow 2^-} f(x) = -(2+1)^2 + 7 = -2$

$\lim_{x \rightarrow 2^+} f(x) = \frac{2}{2} - 6 = -5$

$\lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$
 $\therefore \lim_{x \rightarrow 2} f(x) = \text{DNE}$

c) $f(x)$ is continuous throughout domain except:

$f(x)$ has jump discontinuity at $x=2$
because $\lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$ and
 $\therefore \lim_{x \rightarrow 2} f(x) = \text{DNE}$.

$f(x)$ also has jump discontinuity at $x=6$
because $\lim_{x \rightarrow 6^-} f(x) \neq \lim_{x \rightarrow 6^+} f(x)$ and
 $\therefore \lim_{x \rightarrow 6} f(x) = \text{DNE}$.

$\lim_{x \rightarrow 6^-} f(x) = \frac{6}{2} - 6 = -3$

$\lim_{x \rightarrow 6^+} f(x) = -4$